NEXUS · DOCUMENTATION PACKAGE · DELIVERABLE 06

Implementation Plan

Completed work with evidence, and remaining work ordered by dependency with priorities, estimates, acceptance criteria and tests.

Document control

Document ID	NX-06-PLAN
Version	2.0
Date	2026-11-05
Status	Draft for team review (not approved)
Intended readers	Development team planning the final PoE; assessor reviewing project management
Evidence baseline	Branch New-Changes: commit 716bedb plus the uncommitted Part 3 working-tree changes of 2026-11-05 (v1.0 analysed 3a3f275)
Prepared by	AI-assisted (Claude Code) from repository evidence; team review pending
Approved by	Unknown No approval recorded

Evidence labels used in this document: Implemented verified in code, configuration or test output; Inferred reasoned from evidence but not executed; Recommendation proposed, not implemented; Open question needs a decision; Unknown no evidence available; Estimate planning figure, not a measurement.

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1 Completed work

This history is reconstructed from `PROJECT_IMPROVEMENTS.md`, merged pull requests and commit messages. Dates are commit or merge dates **Implemented**.

Delivered increments

Date	Increment	Evidence	Requirements
2026-08-23	Initial Android app with auth integration, secure token storage, nav flow, splash animation	Commits 65385b2, 0496717, 964cbf5	FR-01, FR-02, FR-19
2026-09-22	CI workflow, Room read cache, task due date / priority / status / labels / checklist, dashboard insights, auth validation	PRs #10–#16; commit 8b31aa3	FR-09, FR-11
2026-09-23	Backend remediation: mutex + atomic writes, Zod validation, JWT secret check, bcrypt, rate limiting, health check, Pino; task detail screen	PR #23; `server.test.js`	NFR-01, NFR-04, FR-12
2026-09-24	Android structural remediation: six repositories, manual DI, `HabitScoring` domain, feature packages, type-safe routes, shared fakes	PR #24	TR-01 to TR-03, FR-14
2026-09-26	UI refinement: theme colours, logo component	PR #25	NX-04-DES
2026-11-05	**Part 3 / v2.0:** Google SSO, FCM push, WorkManager reminders, full zu/tn localisation, audit fixes (AUD-01/02/03/05/06/10/11/16/17/18/21), Settings crash fix, release signing, Play Store kit, recaptured screenshots	`716bedb` + uncommitted working tree	FR-21, FR-22, FR-23, FR-18
2026-10-04	Offline upload queue and `/api/sync`, sync banner, conflict review, project search, task filters, zu/tn resources, configurable `NEXUS_API_URL`, HTTPS-only release, Dockerfile/compose	PR #26 (commit 3a3f275)	FR-16, FR-17, FR-18, TR-07

2 Remaining work

Tasks are ordered so that each depends only on tasks above it. Priority: **P0** blocks a credible release or submission, **P1** is needed before real users, **P2** is improvement. All effort figures are **Estimate** person-day estimates by the document author, not team commitments, and should be re-estimated by the team.

Dependency-ordered backlog

ID	Task	Pri.	Depends on	Est. (days)	Acceptance criteria	Tests
IMP-01	Restore GitHub Actions execution (resolve the account billing lock shown in run annotations) and re-run CI on `main`	P0	None	0.25 **Estimate**	Android CI and Backend CI both green on `main` at or after 3a3f275	Existing CI jobs (CI-01, CI-02)
IMP-02	Fix Zod 4 compatibility before merging PR #27: use `error.issues`; give `z.record` a key schema	P0	IMP-01	0.5 **Estimate**	Backend tests pass on Zod 4; validation still returns 400 `VALIDATION_ERROR`	`npm test`; existing Zod tests

ID	Task	Pri.	Depends on	Est. (days)	Acceptance criteria	Tests
IMP-03	Fail safe on an unreadable data file: refuse to start (or start read-only) instead of starting empty	P0	None	0.5 Estimate	A corrupt `nexus-data.json` stops the process with a clear log line and leaves the file untouched	New Jest test with a corrupt fixture (AUD-01)
IMP-04	Correct the auth error messages: 401 at login gives "Email or password is incorrect"; 409 gives "Email already registered"; 429 gives "Too many attempts"	P0	None	0.5 Estimate	Each status shows its own message; session expiry still shows the expiry text elsewhere	`AuthViewModelTest` cases per status (AUD-18)
IMP-05	Remove or fix the root `docker-compose.yml`; document `backend/compose.yaml` as the single container path	P1	None	0.25 Estimate	`docker compose up` from the documented path persists data across restarts	Backend CI smoke test plus a manual restart check (AUD-02)
IMP-06	Add `app.set("trust proxy", 1)` (configurable) for TLS-proxy hosting	P1	IMP-05	0.25 Estimate	The rate limit keys on the client IP behind the chosen proxy	Supertest with `X-Forwarded-For` (AUD-06)
IMP-07	Provision the hosted API and release signing; set the `NEXUS_API_URL` repository variable	P1	IMP-01, IMP-03, IMP-05, IMP-06	1–2 Estimate	`/health` reachable over HTTPS; signed release APK/AAB produced by CI	Release checklist in NX-07-CICD (AUD-08)
IMP-08	Due-date reminders, or hide the switch (decision OQ-01)	P1	OQ-01	0.25 (hide) / 3–5 (build) Estimate	The switch either has a visible effect or is not shown	Unit test on the scheduler, or UI test that the switch is absent (AUD-10)
IMP-09	Exclude the Room database from backup, or document the choice	P1	None	0.25 Estimate	`backup_rules.xml` and `data_extraction_rules.xml` exclude `nexus_database`	Manual `adb shell bmgr` check (AUD-05)
IMP-10	Enforce lint in CI (`ktlintCheck`, `detekt`) and fix the ktlint parse failure; delete the placeholder `blank.yml` and `build-verification.log`	P1	IMP-01	1 Estimate	CI fails on lint violations; the log file is no longer tracked	CI run (AUD-13)
IMP-11	Accessibility pass: TalkBack walkthrough, contrast measurement, 48 dp targets	P1	None	1–2 Estimate	Results recorded; issues fixed or logged	Accessibility Scanner report; Compose semantics tests

ID	Task	Pri.	Depends on	Est. (days)	Acceptance criteria	Tests
IMP-12	Undo for list deletes of projects, tasks and habits	P2	None	1 Estimate	Snackbar "Undo" restores the item before upload	ViewModel tests
IMP-13	CSV export escaping (RFC 4180 quoting)	P2	None	0.25 Estimate	A habit named `Read, write` exports as one quoted field	`SettingsViewModelTest` (AUD-11)
IMP-14	Fix the habit completion-rate formula: count habit-date pairs	P2	None	0.25 Estimate	Two daily habits both done for 30 days gives 100%	`HabitScoringTest` (AUD-17)
IMP-15	Enable Room `exportSchema` and add `MigrationTestHelper` tests for 3→4→5	P2	None	1 Estimate	Schema JSON committed; migration tests pass	Instrumented tests (AUD-12)
IMP-16	Documentation corrections: README contents links, admin role label, translation claim, recaptured screenshots	P2	IMP-07	0.5 Estimate	Every README link resolves; screenshots match their file names	Link check (AUD-09, AUD-21)
IMP-17	Complete feature-screen localisation (zu/tn)	P2	None	3–5 Estimate	No hard-coded user-facing English strings remain in `ui/`	Lint `HardcodedText`; pseudo-locale run (FR-18)
IMP-18	Final-PoE features (Google SSO, push notifications), as decided in OQ-04	P2	IMP-07, OQ-04	Unknown	Defined per feature in `docs/POE_PART_2_ADDENDUM.md`	Server and end-to-end tests required by the addendum

Total for P0 and P1 excluding IMP-08 "build" and IMP-18: about 5–7 person-days Estimate.

3 Definition of done

Adopted from the quality checklist in `PROJECT_IMPROVEMENTS.md`: API models and validation first; a Room migration for any schema change; loading, offline, retry, unauthorised and validation states covered; ViewModel unit tests plus instrumented coverage of the main path; accessibility checked; README and the improvements log updated Implemented. Recommendation Add "CI green on the PR" once IMP-01 is done.

4 Status at v2.0 (2026-11-05)

Backlog status after the Part 3 work

Task	Status	Evidence or remaining action
IMP-01	Open (team)	Needs the GitHub account billing lock resolved; then re-run CI. Workflows moved to Node 22.
IMP-02	Done	Zod 4.6.7 adopted; 29/29 backend tests pass. PR #27 can be closed as superseded.
IMP-03	Done	Refuses to start; `.bak` copy kept; 3 tests in `part3.test.js`.

Task	Status	Evidence or remaining action
IMP-04	Done	`AuthErrorMessagesTest` ; verified on device.
IMP-05	Done	Root `docker-compose.yml` uses a named volume at `/data` ; `docker compose config` valid. Not run (no Docker daemon).
IMP-06	Done	`TRUST_PROXY` env; not covered by a test.
IMP-07	Partly done	Signing config, upload key and Play kit done. API hosting and Play Console upload are team actions.
IMP-08	Done (built)	Reminders and push implemented (FR-22, FR-23).
IMP-09	Done	Backup rules exclude `database` .
IMP-10	Done (ktlint)	Codebase brought to zero ktlint violations (592 at the start: auto-format, line wrapping, constant renames, and two rules disabled in `.editorconfig` with recorded reasons). `ktlintCheck` now runs in Android CI before the tests. `blank.yml` and `build-verification.log` removed. detekt is not gated because the tool crashes in this toolchain (AUD-23).
IMP-11	Open	No TalkBack or contrast audit recorded.
IMP-12	Done (alternative)	Every delete (project, task, habit) now asks for confirmation, and the project dialog warns that its tasks are deleted too. Verified on device (screenshot 21). Undo was not implemented.
IMP-13	Done	`CsvRowTest` .
IMP-14	Done	New `HabitScoringTest` case.
IMP-15	Open	Room schema export and migration tests not added.
IMP-16	Done	README rewritten (contents links fixed, release notes, AI use); screenshots recaptured.
IMP-17	Done	210 strings × 3 languages; native-speaker review still recommended.
IMP-18	Done (code)	SSO and push implemented; need Google Cloud and Firebase projects to run end to end.